



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads -
AAIB-style causal and contributory factors report

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

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Synopsis

On 07/07/2026 the 24.4 m catamaran passenger ferry MV Tarrakin Star was inbound on TF18, the day's last service to the town wharf at Kanooka Heads, carrying 38 passengers, a master, a deckhand and a kiosk attendant. In the Weerona Channel it held the charted centreline until the chart plotter display froze irrecoverably at 18:07:06. The master, alone on the bridge, continued at about 8 kt rather than stopping, seeking the No. 4 beacon while heading was increased from 195 to 200 degrees and held. The vessel left the dredged channel at about 18:09. The echo sounder shallow alarm sounded at 18:11:48 and at 18:12:04 the vessel struck Boolarong Bank at 6.8 kt, coming to rest on a falling tide 74 m west of the centreline and 54 m outside the channel. One passenger reported a grazed shin and two were treated at the wharf for distress, with no serious injuries. The starboard hull was split in way of the progressively flooding No. 2 void, and the starboard propeller, shaft, rudder stock and skeg damaged. All passengers were ashore by 19:26. Refloated at 00:20, the vessel was withdrawn from service and the channel closed to commercial traffic for 19 hours.

The winter timetable had moved TF18 into darkness without reopening a route risk assessment restricting the channel to daylight transits. This left no night passage plan or hold point for loss of the primary means of position fixing. The plotter had frozen twice that morning, but the fault was passed verbally at the change of master, not entered in the defect book, as handover practice required nothing written, so a firmware fix issued in 11/2024 had never been applied. This practice likely contributed, although an entry that afternoon might not alone have stopped the evening service.

The beacon, the secondary means of fixing, was very likely not visible over that stretch. Mangrove canopy on Weerona Islet, left untrimmed as unfunded for three works rounds, screened it, and a solar conversion not notified to operators had halved its nominal range. A national code imposing no availability, recording or notification duty on aids outside declared shipping channels likely lay behind both. The 0.9 m shallow alarm setting gave 16 seconds' warning against a 92 m stopping distance. Whether new foreshore floodlighting impaired the beacon's detection was undetermined. Steering or propulsion failure and low water were ruled out. The forward saloon lifejacket locker arrangement and the winter roster were unlikely to have influenced the occurrence but increased risk, while the master's decision not to attempt to refloat the holed vessel reduced risk.

Subsequently, a local notice to waterway users gives the beacon's reduced range, the aids to navigation register has been opened to operators and the passage plan reissued. The canopy was trimmed on 24/07/2026, with a three-yearly cycle programmed. Recommendations were made to replace the services agreement's vegetation scope with a sightline test and for Tarrakin Ferries to write a night passage plan with a hold point before night transits resume.

Factual information

History of the occurrence

MV Tarrakin Star was operating service TF18 on 07/07/2026, the last service of the day from Muntari Island to the town wharf at Kanooka Heads. Since 01/06/2026 the winter timetable's evening returns had been made in darkness.

The master took over from the morning master at the town wharf at about 13:05. Both masters stated that the outgoing master told him the chart plotter had frozen twice that morning and recovered by itself both times, and that the relieving master replied it had done the same to him the month before. Neither entered the fault in the wheelhouse defect book, which held no entry of any kind for 07/07/2026 before the occurrence. The master and the deckhand described the afternoon's two round trips as normal.

TF18 departed Muntari Island at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant on board, and the master reported the departure to Port Control on VHF channel 12. Sunset was at 17:02 and civil twilight ended at 17:29, so the channel transit was made in full darkness.

The vessel entered the Weerona Channel from the north at 18:03:58, abeam the No. 1 beacon. At 18:04:14 the master sent the deckhand to collect fares, leaving himself alone on the bridge. At 18:05:30 the vessel was abeam the No. 2 beacon on the charted centreline at 8.4 kt.

At 18:07:06 the chart plotter display froze. The chart picture remained, the vessel symbol was stationary and the display did not accept touch input. Unlike the morning freezes, it did not recover. The master tried the display, then continued at the same speed, looking for the No. 4 starboard-hand beacon on the western side of the channel. At 18:09:40 he slid open a wheelhouse window. In his account, when he sighted the beacon it was dead ahead instead of on the starboard bow, and he knew at once that he was on the bank side. He also stated that he had eased the vessel about five degrees to starboard while searching, without having decided to do so.

The vessel crossed the western edge of the dredged channel at about 18:09 and at 18:12:04 struck Boolarong Bank at 6.8 kt over the ground. It came to rest within 11 m, heeled 4 degrees to starboard, 74 m west of the charted centreline and 54 m outside the dredged channel, on a falling tide.

The master stopped both engines at 18:12:31 and did not restart them. At 18:12:40 he told passengers over the public address that the vessel was stopped and safe. At 18:13:04 he broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16 giving the position, persons on board, no injuries and no pollution, and stated that he was not attempting to refloat. Port Control acknowledged at 18:13:29 and called out the workboat. The master later explained his decision by reference to the falling tide, the unknown bottom, the number on board and the nine seconds of grinding.

The deckhand issued lifejackets from aft forward. In the aft saloon passengers opened the locker themselves. In the forward saloon the locker is behind the kiosk counter, where three passengers were standing, and the deckhand had to pass behind the counter and hand jackets out over it. All passengers were in lifejackets by about 18:22. The deckhand sounded the No. 2 starboard void at 18:16, found approximately 75 mm of water, and sounded it every ten minutes thereafter. The workboat came alongside at 18:41 and the last passenger was ashore at 19:26. The vessel refloated on the rising tide at 00:20 on 08/07/2026 and was towed to the town wharf.

Injuries and damage

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress.

- The starboard hull bottom plating was set up over 4.1 m between frames 6 and 11, with a 180 mm split in way of the No. 2 void, which had flooded to approximately 260 mm by refloating.
- All three starboard propeller blades were bent aft and the starboard shaft was bent 1.4 mm.
- The starboard rudder stock was distorted 3 degrees and the starboard skeg was torn away.
- No damage was found to the port hull, propulsion or steering.

The vessel was withdrawn from service and the Weerona Channel was closed to commercial traffic for 19 hours.

Recorded information

The shore AIS receiver log, engine monitoring recorder, echo sounder alarm log, plotter event log, and VHF and bridge camera audio were recovered. Times were recorded in UTC and are given in local time (UTC +10:00). The plotter and echo sounder logs were corrected by 2 seconds to the AIS receiver clock.

The AIS track showed the vessel within 3 m of the centreline until 18:07:06. The heading then increased steadily from 195 to 200 degrees over about a minute and was held. The rate of turn peaked at 2.1 degrees per minute at 18:07:40 and returned to zero by 18:08:50. The vessel set west at an average of 12 m per minute: 12 m west at 18:08:31, 20 m west at the channel edge at 18:09:00, 39 m west at 18:10:02 and over the bank at 18:11:00. The shallow alarm sounded at 18:11:48 with the vessel 70 m west. Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement.

The engine monitoring recorder showed both engines at 1,780 and 1,784 rpm from 18:03 to 18:11:52, with no alarm or temperature or pressure excursion. Steering hydraulic pressure was 96 to 99 bar against a normal band of 90 to 105 bar, and rudder angle followed helm demand within 1.5 degrees throughout. At 18:11:52 both throttles moved to the astern detent together. Astern thrust was developing at contact but had not reached full.

The bridge audio recorded the master saying "Oh, come on" at 18:07:11, with sounds consistent with repeated screen taps at 18:07:26. At 18:08:31 he said "Right. Where's four", and at 18:10:02 "There's four. Keep her over". The shallow alarm tone ran from 18:11:48 to 18:12:20. A heavy impact at 18:12:04 was followed by about 9 seconds of grinding and vibration.

The vessel was 0.68 nm north of the No. 4 beacon when the plotter froze, 0.47 nm when the master began looking for it and 0.25 nm when he sighted it.

Equipment information

MV Tarrakin Star is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers. It completed its annual survey on 11/03/2026 with no outstanding conditions and was last slipped in 03/2025. The defect book held eleven entries in the preceding 90 days, none concerning navigation, steering or propulsion. The maintenance system held no work order, fault report or correspondence on the chart plotter in 2026.

The chart plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024. Its event log recorded watchdog resets with cause code 0x51 at 10:42 and 11:17 that morning, and at 18:07:06 recorded that the display

process was not responding, with no reset. The operator did not hold the manufacturer's firmware bulletin of 11/2024 addressing cause code 0x51 and had no subscription to the manufacturer's bulletins. The outgoing master believed the plotter had frozen three or four times on this vessel in the past year, none reported.

The echo sounder shallow alarm was set at 0.9 m under the keel, the value entered at commissioning in 2019. The trials crash stop distance from 8 kt is 92 m. The vessel covered 58 m between the alarm and contact.

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon locker holds 40 adult jackets in the bulkhead behind a kiosk server counter fitted in a 2019 refit under a minor modification notice.

Meteorology

The master described the conditions as clear, with a light westerly and no sea in the entrance.

Personnel information

The master had 14 years as a master, 11 on this route, and approximately 4,100 transits of the Weerona Channel. He had held a certificate of competency as master less than 35 m since 2012, and his last route check, on 14/02/2026, was satisfactory. He was on day nine of a twelve-day tour, on a split duty of 05:40 to 08:40 and 13:10 to 19:20, and had slept 6 h 15 the night before. He described himself as tired, normal for a split, and not slow.

The outgoing master had 9 years as a master, 4 on this route, with approximately 1,400 channel transits. He had worked 05:40 to 13:05 on day six of a twelve-day tour.

The deckhand had 7 months with the operator and was in her first winter season on the route. She stated that she had never been shown the channel marks.

Organisational and management information

The passage plan appendix named a speed, a course and three beacons, recorded the No. 4 beacon at a range of 6 M, and did not address who was to be on the bridge. There was no night passage plan for the channel. Both masters and the deckhand stated that the winter timetable change came with a new roster sheet and nothing else.

Both masters described the practice of passing on verbally a fault that had cleared, without entering it in the defect book. The outgoing master stated that, had the plotter frozen in the channel, he would have taken the way off and waited, and that he knew of no rule requiring this. The operator's reporting system has a "navigation hazard" category, unused on this route in three years, and no master interviewed knew it existed. Both masters stated that the No. 4 beacon had become harder to see over two years and that neither had reported it. The master stated, "There's no form for the beacon looks dim."

Tests and research

On 12/07/2026 the steering gear rams, tiller arms, hydraulic system and autopilot were removed and function tested. All were serviceable and the system passed a lock-to-lock timing test. The chart plotter was bench tested for 46 hours from 21/07/2026 with the same antenna and cabling, and the fault did not recur. On 16/07/2026 the forward saloon locker face was found to be unreachable from 22 of the 34 seats without passing behind the counter through a 0.6 m gap. The muster placard directs passengers to a locker not visible from the seats. The same counter is fitted to the operator's two other 24 m vessels.

Other information

The survey of the grounding position on 08/07/2026 found the bed at the contact point 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west, a gradient of about 1 in 70. Furrows ran 11 m from first contact on a heading of 197 degrees. The contact point was 34 m west of the No. 4 beacon.

The No. 4 beacon (Fl G 2.5s) was converted from mains supply to a solar lantern in 11/2025, and its nominal range fell from 6 to 3 nautical miles. The hydrographic service was notified on 04/02/2026 and the chart correction was pending on the day. No local notice to mariners was issued and no operator was told. A check on 18/07/2026 measured a luminous range of 2.6 nm and found the light not visible from the channel between 0.84 nm and 0.34 nm north of the beacon, because the sightline passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy measured 5.8 m against a lantern height of 4.2 m above chart datum.

A work order to trim the canopy was raised in 09/2023 and deferred as unfunded at the 2023, 2024 and 2025 works rounds. A further order in 03/2026 recorded the sightline as fully closed. The islet falls outside the 50 m scope of schedule 2.4 of the services agreement. The monthly check, made from the town wharf, recorded the beacon as operating every month from 01/2024. No defect or notice had ever been raised against the beacon, and the aids to navigation register had never been provided to Tarrakin Ferries. Both masters referred to new car park floodlighting behind the beacon. The master stated that he could not say whether the lighting was why he could not see the beacon.

Analysis

External Influences

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

The aids to navigation in the Weerona Channel are governed by Waterways Aids to Navigation Code WAN-2. Section 4.1 requires the responsible authority to maintain an aid "so that it performs the function for which it was established". The more demanding duties sit in section 4.2, and they apply only "where an aid to navigation is established on a declared shipping channel". On such a channel the authority must record the aid's availability, must achieve "an availability of not less than 99.0 per cent measured annually", and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect.

The Weerona Channel is not a declared shipping channel. It is a dredged recreational and commercial access channel on a waterway managed by Kanooka Shire Council through the Kanooka Heads Port Authority. Section 4.3 therefore applies to all six of its beacons. It requires only that the authority "be able to demonstrate that the aid is inspected at intervals it has determined and recorded." The services agreement between the Council and the Authority sets those intervals. Schedule S2.1 provides for a daylight inspection once per quarter and a monthly confirmation of operation "by observation from the town wharf". Schedule S2.4 places vegetation beyond 50 m of an aid outside the agreement's scope. Schedule S2.7 states that there is "no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." The text of the code, the agreement and the channel's status are documentary and not in dispute. The Authority was not in breach of either instrument. It is virtually certain that this regime existed as described.

Whether the regime influenced the occurrence was a more finely balanced question. The case for influence is that each mechanism the code provides for carrying an aid's condition to its users stops at the boundary of this channel. These mechanisms are a recorded availability figure, a measured performance standard and a notice to mariners of a change in range. The No. 4 beacon was converted to a solar lantern in 11/2025, and its nominal range fell from 6 nautical miles to 3. A notification was lodged with the hydrographic service, but the chart correction was still pending on the night. No local notice to mariners was issued and no operator was told. On the night of the occurrence, the operator's passage plan appendix still recorded the beacon at 6 M. Had the notification duty at section 4.2 applied, it would have been the one mechanism that put the range change in front of the operator before the change took effect. The regime also set the limits within which the Authority looked at the beacon. The monthly confirmation was made from the town wharf, 2.1 nm south of the beacon. Weerona Islet, which screens the light from the northern approach, is not in the line of sight from that position. The beacon was recorded as observed and operating in every month from 01/2024 to 07/2026. The islet stands 240 m from the beacon, so clearing its canopy fell outside the 50 m vegetation scope of the agreement.

The argument against influence was also substantial, and the investigation gave it weight. Nothing in the code prevented the Authority from notifying operators, trimming the mangroves or measuring the light. The evidence indicates that these things were not done for reasons of funding and priority, not because the code permitted their omission. Moreover, an availability target of 99.0 per cent would not have detected the condition that mattered. The lantern was lit and working throughout. The difficulty was that it could not be seen from the channel between 0.84 nm and 0.34 nm north of the beacon, and no availability figure measures that. On this reading, the limited scope of the code was background to the Authority's decisions, not a cause of the occurrence.

The investigation weighed these arguments and concluded that the regime likely influenced the occurrence. Two points decided it. First, a notification duty would have required the operator to be told of the range change. Second, the operator's passage plan still carried the superseded figure on the night, which shows that no other route had brought the change to its attention. This conclusion is less firmly established than the other influence findings in this report. It rests on the balance of the evidence, and the counter-argument above was not eliminated.

The regime's effect on the occurrence ran through two decisions at the Authority. One was the solar conversion that halved the beacon's nominal range without any operator being told. The other was the deferral of the work to clear the beacon's sightline, unfunded, at three successive works rounds. Both were found to be causal factors, and they are analysed under Organisational Influences. They led in turn to the No. 4 beacon not being visible from the channel over the half mile before the vessel left it, which is analysed under Local Conditions.

Because the investigation determined that this factor influenced the occurrence, it is a causal factor in this report, not a contributory one. The qualification on that determination carries through to the findings. It is virtually certain that the beacons on the Weerona Channel were subject to no availability standard and no duty to notify users of a change, and it is likely that this contributed to the grounding through the decisions it left to the Authority's discretion.

Organisational Influences

Winter timetable moved the last service into darkness without reopening the route risk assessment

The route risk assessment for the Weerona Channel, RRA-07 issue 2 of 03/2023, was the current issue, and no issue 3 existed. It named, among its hazards, the "single unlit reach between the No. 2 and No. 4 beacons". It carried an explicit restriction: "daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan."

Change record CHG-2026-014 moved service TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026. It described the change as "timetable change only" and recorded "No new hazard identified". Against the question of whether a route risk assessment review was required, it answered "No". The Operations Manager approved it on 22/04/2026, one level below the post the restriction names. The change record did not refer to the daylight restriction. RRA-07 was not reopened, and no night passage plan was written. The General Manager Operations' approval was neither sought nor given. From 01/06/2026 every TF18 transit of the channel was made after the end of civil twilight, and the service had run 31 times in darkness before the occurrence. Both documents are current controlled copies, and it is virtually certain that the change was made in this way.

The change process asked whether the change introduced anything new. It did not ask whether the existing assessment still held. A review of RRA-07 against a night operation would have reached the unlit reach the assessment already identified, and the restriction itself would have required a specific passage plan before the first service ran. The consequence of that omission was the absence of any night passage plan, and of any hold point should the primary means of fixing be lost. That absence is argued in the analysis of risk controls.

Two matters limit the strength of the link. First, a review might have produced a plan that said no more than the daylight plan does. Second, what the Operations Manager understood the restriction to require was not established, so the reason the change was assessed as it was remains unclear. These matters go to how the change came to be approved, not to whether it was approved.

It is likely that moving TF18 into darkness without reopening RRA-07 contributed to the occurrence, and it is a causal factor.

Beacon converted to solar, nominal range halved, and no operator was told

In 11/2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern was rated at a nominal 3 nautical miles, against 6 for the lantern it replaced, and the character, Fl G 2.5s, was unchanged. A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued, and no operator using the channel was told. The range check made from the port workboat on 18/07/2026 measured a luminous range of 2.6 nm. Each of these points is documentary or independently measured, and it is virtually certain that the condition existed.

The master was working to Tarrakin Ferries' passage plan, OPS-3 Appendix A, issue 2. That plan listed "No. 4 (Fl G 2.5s, range 6 M)" and named the "visual transit of the No. 4 beacon" as the secondary means of fixing. The written plan therefore described a light more than twice as strong as the one actually exhibited. By the time of the occurrence the lantern had been a 3 M unit for about eight months. A weaker light than expected is harder to acquire, and the master searched for it for 91 seconds before calling it at 0.25 nm.

The evidence against a stronger link is that the whole of that search took place between 0.84 nm and 0.34 nm north of the beacon. Over that stretch the light was screened entirely. A screened light is not made visible by being brighter. This is why the link is put no higher than likely, and the effect of the reduced range is confined to the conditions in which the master was trying to acquire the light. The possibility that new foreshore lighting behind the beacon also affected its detection was examined separately, could not be resolved, and is not relied on here.

It is likely that the unnotified halving of the No. 4 beacon's nominal range contributed to the occurrence, and it is a causal factor.

Clearing of the beacon's sightline deferred unfunded for three works rounds

The Port Authority's aids to navigation register showed that the screening of the No. 4 beacon was known for nearly three years before the occurrence. Work order WO-118, raised in 09/2023, identified mangrove regrowth on Weerona Islet screening the beacon from the northern approach. It recommended trimming to 2 m on a three-yearly cycle, at an estimated \$18,500. The work was deferred as unfunded at the 2023, 2024 and 2025 works rounds. Work order WO-176, raised in 03/2026, recorded the sightline over the northern approach as fully closed.

The Authority's file note against WO-176 explained the funding position. Because the islet is 240 m from the beacon, it fell outside the 50 m scope of schedule 2.4 of the services agreement. The work therefore competed with boat ramps and pontoons in the general waterways vote rather than with aids to navigation. The canopy, given as 2.1 m in the 2019 aerial survey, measured 5.8 m on 18/07/2026, against a lantern height of 4.2 m above chart datum. The condition was documented by the Authority itself and confirmed on site, and it is virtually certain that it existed.

The range and visibility check located the screened arc between 0.84 nm and 0.34 nm north of the beacon. The vessel crossed the channel's western edge at about 18:09, 0.40 nm north of the beacon, inside that arc. The master began looking for the light at 0.47 nm and called it at 0.25 nm, shortly after the vessel emerged from the arc. Over the stretch in which the secondary means of fixing was needed, the light was not dim or defective but absent. The only reservation is that the master might not have picked up the light in that minute and a half even had it been unscreened.

It is very likely that the repeated deferral of the sightline clearing contributed to the occurrence, by leaving the No. 4 beacon invisible over the stretch in which the vessel left the channel, and it is a causal factor.

Risk Controls

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The transit was made under OPS-3 and its appendix A, issue 2, which was the current issue. Section 3.4 required the transit to be made "on the charted centreline, monitored on the chart plotter". The appendix named the chart plotter as the primary means of position fixing and gave as the secondary "visual transit of the No. 4 beacon and the town wharf leading lights." Its only reference to darkness was the closing note: "Night transits are not scheduled on this route." It is virtually certain that no night passage plan existed for the Weerona Channel. The documents establish this, and the operator confirmed it.

The plan had a further gap that mattered more on the night. It gave no guidance on what to do if the plotter was lost during the transit, and it named no abort or hold point anywhere. The secondary means offered less than its wording suggested. The town wharf leading lights lay 2.1 nm beyond the No. 6 beacon and did not resolve as two lights until the vessel was inside the channel's southern third. Where the plotter froze, 0.68 nm north of the No. 4 beacon, the secondary means therefore consisted of the beacon alone. Section 3.6, which required a reduction in speed and consideration of anchoring, applied only "in reduced visibility". The visibility was not reduced, so 3.6 did not apply.

The plotter froze at 18:07:06, and the vessel crossed the channel's western edge at about 18:09. This left the master 114 seconds in which stopping was available, on flat water with no other traffic. A named hold point would have applied throughout that interval. Both masters interviewed said independently that the right response was to stop, and both said that nothing in the operator's procedures called for it. Some doubt remains. A master who did not think of stopping unprompted might not have applied a written rule under the same pressure. Even allowing for that, it is very likely that the absence of a hold point contributed to the decision to continue the transit at speed. That decision is argued under Individual Actions. This gap stemmed from the winter timetable change, which moved the last service into darkness without reopening the route risk assessment, as set out under Organisational Influences.

The absence of a night passage plan and of a hold point on loss of the primary means of fixing was a causal factor.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

OPS-9 9.1 required immediate reporting only of a defect that rendered the vessel unseaworthy or affected a statutory item. The chart plotter was not a statutory item on this class 1D vessel, although the passage plan named it as the primary means of position fixing. Section 9.2 placed all other defects in the defect book "at the end of the shift in which they are found". For a fault that clears itself, that requirement provides no reliable trigger. Section 9.3 required the outgoing master to brief the relieving master on "any defect". Nothing required that brief to be written down, and nothing required the relieving master to confirm that a defect had been entered. Both masters independently described the same practice: the book recorded what was broken at handover, and faults that had cleared were passed on by word of mouth. It is virtually certain that this arrangement existed.

The plotter's event log recorded freezes at 10:42 and 11:17 that morning. The defect book held no entry for 07/07/2026 before the occurrence. The maintenance system held no plotter work order, fault report or correspondence at any time in 2026. The outgoing master believed the plotter had frozen three or four times in the previous year and that none of these freezes had been written up. The firmware issued in 11/2024, which addressed the fault's cause code, had not been applied. The operator held no subscription to the manufacturer's bulletins.

There were two readings of this evidence. On the first, the handover arrangement produced exactly the outcome it permitted. A recurring fault on the primary means of fixing was known to masters but never reached anyone ashore who could act on it. As a result, the firmware fix was never applied, and the master met the fault on the night with no reason to treat it as more than a nuisance. On the second, an entry made at 13:05 would have been reviewed weekly rather than that afternoon, and TF18 would have sailed either way. On that reading, the arrangement is a genuine weakness across the fleet but not a cause of this grounding. The investigation preferred the first reading because the significant failure was the year of unreported freezes, not the single missing entry. Even so, the influence is put no higher than likely. This explains how the omission at the handover came about. It does not make the omission any less significant.

The verbal change-of-master defect brief, with no check that defects were recorded, was likely a causal factor through the unreported history of the plotter fault.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

The forward saloon locker held 40 adult lifejackets and was mounted behind the kiosk serving counter. The counter was fitted in a 2019 interior refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. The survey of 16/07/2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap. With people standing at the counter, that gap was the only route. The muster placard directed passengers to a locker they could not see from their seats. The arrangement approved at build in 2011 had left the locker face clear. The same counter is fitted to the operator's two other 24 m vessels. It is virtually certain that the condition existed.

The arrangement did not affect the grounding, and it did not affect the outcome. All 38 passengers were in lifejackets by about 18:22 and ashore by 19:26. The vessel was upright and stable throughout. Its influence on this occurrence was unlikely.

Its importance is very likely. If the occurrence had involved a fire, a night evacuation into the water, or an injured deckhand, passengers who could not reach the locker without crew help would have had to wait for jackets. The underlying weakness is the modification route, which allowed an interior refit to alter a safety equipment arrangement without examining it.

The inaccessible forward saloon locker was a contributory factor that increased risk, though not one that influenced this occurrence (Safety issue).

Local Conditions

Chart plotter display froze during the channel transit and did not recover

Three independent sources establish the freeze. The plotter's event log recorded the display process not responding at 18:07:06, with no reset following. The bridge audio has the master saying "Oh, come on" at 18:07:11 and "(unintelligible) frozen again" at 18:07:44, with screen taps in between. The master described the chart picture held on the screen, the vessel symbol stationary and the display not accepting touch input.

The 46 hour bench test on 21/07/2026 did not reproduce the fault and found no defect. That result bears on why the display froze, not on whether it froze. The contractor identified cause code 0x51 as a fault known to the manufacturer. That fault was addressed by firmware issued in 11/2024, which had not been applied to this unit. The same code had been logged at 10:42 and 11:17 that morning, and on both occasions the unit recovered itself.

The track shows the effect. While the plotter worked, the vessel held the centreline to within 3 m. It began to set west from the moment of the freeze and reached the channel's western edge less than two minutes later. The master's search for the No. 4 beacon began 85 seconds after the freeze. From 18:07:06 he was without the primary means of position fixing that the passage plan required. How the morning's freezes were passed on, and why the transit continued at 8 kt, are considered elsewhere in this analysis.

It was virtually certain that the display froze and did not recover, and extremely likely that this influenced the grounding, which makes it a causal factor.

No. 4 beacon not visible from the channel over the half mile before the vessel left it

The range and visibility check of 18/07/2026 was made from the port workboat, on the inbound track and at the same time of night. It found that the beacon cannot be seen from the channel between 0.84 nm and 0.34 nm north of it. Over that stretch the sightline passes through the mangrove canopy on Weerona Islet, which was measured at 5.8 m against a lantern height of 4.2 m.

The vessel was 0.68 nm north of the beacon when the plotter froze. It was 0.47 nm north when the master said "Right. Where's four" and 0.25 nm north when he said "There's four. Keep her over", 91 seconds later. The whole of his search therefore took place inside the screened arc. During it he slid open the wheelhouse window, in case the glass was the problem. Both masters raised the beacon unprompted in interview.

Two points weigh against the factor, and neither changes the conclusion. First, the measurement was made eleven days after the occurrence, and the canopy was not surveyed on the night. Second, the vessel had begun to set before the master started looking. With the plotter frozen, the beacon was the only means of fixing left. It was absent for the whole period in which the vessel was set west. The reasons the light was screened and dimmed are considered under Organisational Influences.

It was extremely likely that the beacon was not visible over that half mile, and very likely that this influenced the grounding, which makes it a causal factor.

New foreshore floodlighting behind the beacon reduced the master's ability to pick it up

The investigation examined whether the new boat ramp floodlighting made the beacon harder to detect. The question could not be settled.

The evidence supporting the hypothesis was as follows:

- The master described looking "into a wall of light" and called the beacon "a nothing" against the town and the new car park.
- The outgoing master, independently, called it "washed out altogether from up the top".
- The lighting survey found 14 unshielded LED floodlights on 12 m columns, each presenting about 4,800 candela toward the channel. They were commissioned on 22/05/2026.

- From 0.5 nm north of the beacon, the array lay behind it within 4 degrees of bearing and 1.5 degrees of elevation. The array was continuously lit while the beacon showed a 0.4 second green flash every 2.5 seconds.
- No assessment of the lighting against the channel's aids was required or made.

The evidence against the hypothesis was as follows:

- The only point where the alignment was measured lies inside the screened arc. Over the stretch where the master was looking, the beacon would not have been visible against a black background either.
- The light became visible again with the floodlights unchanged.
- There is no measurement from the night itself.
- Nothing establishes a detection threshold for that flash against that background.

The alignment is real and the masters' accounts agree. However, the evidence cannot separate the effect of the floodlights from that of the mangroves. That question may not be answerable while the canopy screens the light. It was about as likely as not that the floodlighting reduced the master's ability to detect the beacon. The matter remains unresolved and supports no finding.

Water level below prediction reduced the depth over Boolarong Bank

This hypothesis was ruled out as very unlikely. The Port Authority's gauge read 0.91 m at 18:12 against a predicted 0.94 m, so the water level did not explain the grounding. The vessel touched because it was 54 m outside the dredged channel.

Day nine of twelve, on a split duty across a thirteen hour forty spread

The duty pattern is documented. 07/07/2026 was the master's ninth consecutive day. The duty was 9 hours 10 minutes across a spread of 13 hours 40 minutes, inside both the operator's limits and the state crewing determination. The master recorded 6 hours 15 minutes of sleep in a fatigue diary that the operator does not collect. He said in interview: "the roster's been like that for eleven years and I knew where I was."

The recorded evidence showed no sign of degraded performance:

- The helm held the centreline to within 3 m.
- The response to the shallow alarm took 4 seconds, against a fleet figure of 6 to 9 seconds.
- The radio work was correct at the first attempt.
- The decision not to refloat was sound.

The one decision fatigue could have affected was whether to stop. Both masters said independently that no document prompts that decision. It was therefore unlikely that fatigue influenced this occurrence. It remains likely to matter on the route, because the exposure recurs on every darkness transit of the winter timetable. It is reported as a factor that increased risk, not as part of this grounding.

Individual Actions

Plotter fault passed on verbally at the change of master and not entered in the defect book

That the fault was passed on but not written down is virtually certain. Four sources agree on it: the defect book, the maintenance system and the accounts of both masters. The plotter's event log shows two freezes during the morning shift, at 10:42 and 11:17, and the unit recovered from both by itself. At the change of master at about 13:05, the outgoing master told the relieving master

about them. The relieving master replied that the plotter had done the same to him the month before. Neither made an entry in the wheelhouse defect book, and neither reported the fault to the Operations Manager. No entry of any kind was made for 07/07/2026 before the occurrence. Both accounts of the handover agree in every particular. Both masters raised the omission themselves before they were asked about it.

Nothing required either master to do otherwise. OPS-9 at 9.2 provides that a defect on a non-statutory item "shall be entered in the vessel defect book at the end of the shift in which they are found". The chart plotter is not a statutory item on this route. OPS-9 at 9.3 requires only that the outgoing master "will brief the relieving master on the state of the vessel, any defect". It does not require the brief to be written, and it does not require the relieving master to confirm that a defect has been booked. The omission is therefore best understood as the ordinary working of that arrangement, not as a departure by either master. The weakness in the arrangement itself is argued under Risk Controls.

The effect of the omission was likely, rather than certain. An entry would have put a recurring fault in front of the people ashore who buy firmware and decide when vessels are withdrawn from service. The maintenance system holds no plotter entry for 2026, including none for the earlier occurrence the relieving master recalled. The operator held no subscription to the manufacturer's bulletins. As a result, firmware issued in 11/2024 that addresses cause code 0x51 had never been applied. Against this, the entry would have been made at about 13:05, and the Operations Manager reviews the defect book weekly. A single entry would not by itself have kept TF18 from sailing with the same unit that evening.

The investigation therefore concluded that the unrecorded plotter fault was a causal factor, which likely allowed a known and already corrected firmware fault to remain on the vessel's primary means of position fixing.

Transit continued at eight knots after the plotter froze, rather than stopping in the channel

That the transit continued at eight knots is virtually certain. The track, the engine recorder, the bridge audio and the master's own account all agree on it. Speed over the ground was held at 8.4 kt until 18:08 after the freeze at 18:07:06. It then fell to 7.6 kt by 18:11, with no throttle movement recorded, which is consistent with drag from shoaling water. At 18:08:31, 85 seconds after the freeze, the master is heard saying "Right. Where's four". From that point his attention was on the search for the beacon. Nothing on the audio indicates that stopping was considered.

In interview and in his safety report, the master said that he did not consider stopping. He said he turned straight to the visual transit because he regarded it as the backup. He also said that, had he been asked beforehand, he would have said he would stop. Conditions did not argue against stopping. There was no other traffic in the channel, the water was flat and the wind was offshore at 8 kt. The vessel's crash stop distance from 8 kt is 92 m. No lookout was on the bridge, because the deckhand had been sent to collect fares at 18:04:14. This was normal practice on the route, and OPS-3 does not address it either way.

The vessel crossed the channel's western edge at about 18:09, 114 seconds after the freeze. A stop at any point in that interval would have left the vessel in dredged water rather than on the bank. On that basis, the continuation of the transit extremely likely contributed to the grounding.

The decision did not arise in isolation. The passage plan names the plotter as primary and the visual transit as secondary, but it contains no hold point and nothing on what to do if the plotter is lost. That gap is argued under Risk Controls. The beacon the master was searching for could not be seen over the stretch in which he looked, and that matter is argued under Local Conditions.

The investigation therefore concluded that continuing at eight knots without a means of fixing position was a causal factor, which extremely likely took the vessel out of the channel and onto Boolarong Bank, and which sat downstream of the passage plan and the screened beacon.

Engines stopped and no attempt made to refloat on a falling tide

That the master stopped the engines and did not attempt to refloat is virtually certain. It is shown by the audio and the engine recorder, and the survey confirms the breach the decision guarded against. Both engines were stopped at 18:12:31, 27 seconds after contact, and were not restarted. At 18:13:04 the master's pan pan stated "I am not attempting to refloat" as a decision already made. He then had the No. 2 starboard void sounded and reported the ingress to Port Control at 18:17:05 as "approximately seventy five millimetres on the sounding rod". He arranged for the void to be sounded every ten minutes thereafter. He addressed the passengers at 18:12:40 and again at 18:14:52, and all 38 were issued lifejackets.

The survey found a 180 mm split in the starboard plating in way of the No. 2 void. Ingress had reached approximately 260 mm when the vessel refloated at 00:20. In interview, the master gave his reasons as the falling tide, the unknown bottom, the number of people aboard, and the fact that the vessel could not sink while it sat on the bank. Backing off into 3.2 m of water with the damage unassessed would have put a flooding compartment into deep water with the passengers still aboard.

The investigation therefore concluded that the decision not to refloat very likely reduced the risk to those on board, and that it was correct and promptly made.

Occurrence Events

Vessel left the dredged channel and tracked onto Boolarong Bank

It is virtually certain that the vessel left the dredged channel and tracked onto Boolarong Bank. Three independent sources fix where the vessel was: the shore AIS track, the surveyed grounding position and the furrows in the bed.

The AIS track shows the vessel 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30. It held the centreline until 18:07:06, which is the time the chart plotter's event log recorded the display freeze. The heading then increased steadily from 195 to 200 degrees over about a minute and was held there. The recorded rate of turn peaked at 2.1 degrees per minute at 18:07:40 and had returned to zero by 18:08:50.

The track then shows the vessel's offset from the centreline growing as follows:

- 12 m at 18:08:31
- 20 m at about 18:09, at the channel's western edge
- 39 m at 18:10:02
- 58 m at about 18:11, over the bank

Contact was made at 18:12:04, 74 m west of the centreline. The survey of 08/07/2026 put the grounding position 54 m outside the channel's western edge and 34 m west of the No. 4 beacon. Furrows in the bed ran 11 m from the point of first contact on a heading of 197 degrees.

The shape of that track matters as much as its end point. The westward movement did not come from current or wind. The wind was offshore at 8 kt, and there is no cross-set in the channel at that state of tide. What the recorder shows is a heading change that was applied over about a minute and then held. That is consistent with a vessel being steered onto a new heading and kept there. It began within a minute of the plotter freezing.

The bridge audio places the master's search for the beacon inside the same interval. He said "Right. Where's four" at 18:08:31 and "There's four. Keep her over" at 18:10:02. By the second remark the vessel was already 39 m west of the centreline and outside the dredged channel.

The reasons the vessel was steered onto that heading are argued elsewhere. The No. 4 beacon was not visible over the stretch in question, and that is addressed under Local Conditions. The continuation of the transit at eight knots after the plotter froze is addressed under Individual Actions. The question of whether the new foreshore floodlighting affected the master's detection of the beacon is unresolved, and it is treated in its own subsection.

The channel is 40 m wide and dredged to 3.2 m. The grounding followed directly from the vessel being outside it, and from nothing else. Had the vessel remained within the channel, it would not have grounded.

The vessel's departure from the dredged channel was virtually certain to have contributed to the grounding and was a causal factor.

Shoaling not detected until sixteen seconds before contact

It is virtually certain that the shoaling went undetected until sixteen seconds before contact. The echo sounder alarm log, the bridge audio and the recorded speed agree to the second.

The echo sounder's shallow alarm was set to 0.9 m under the keel. In the words of the alarm log, this was "the value written into the setup when the unit was commissioned in 2019 and it has not been changed". The alarm sounded once on the voyage, at 18:11:48. Both throttles were moved to astern together four seconds later, at 18:11:52. Contact followed at 18:12:04.

At 7.1 kt the vessel covered 58 m in those sixteen seconds. The vessel's own trials figure for a crash stop from 8 kt is 92 m. Astern thrust was developing at the moment of contact but had not reached full. The master's response to the alarm was therefore prompt, but the distance left to him was shorter than the distance the vessel needed to stop.

The surveyed bank gradient across the contact section was about 1 in 70. On that slope, the alarm log states that the alarm "could not have sounded appreciably earlier at that setting".

The equipment did exactly what it was set to do. The shortfall lay in the combination of three things:

- the 0.9 m setting
- a shallow gradient of about 1 in 70
- a speed of about eight knots

Together these gave less warning than a stopping distance. A setting that gave about forty seconds would very likely have allowed the vessel to stop short of the bank. So would a transit speed at which sixteen seconds was enough to stop.

It is extremely likely that the late detection of the shoaling contributed to the grounding. Because a warning long enough to stop would have prevented the contact, the late detection is a causal factor and not merely a contributory one.

The late detection of the shoaling, at a setting and speed that left 58 m against a 92 m stopping distance, extremely likely contributed to the grounding and was a causal factor.

Steering or propulsion failure caused the vessel to sheer off track

A steering or propulsion failure was ruled out as very unlikely, on two independent grounds:

- The engine monitoring recorder showed both engines within 12 rpm of each other throughout, with no alarm or throttle movement. It showed steering hydraulic pressure continuously within its normal band, rudder angle following helm demand within 1.5

degrees, and a rate of turn that was applied and then returned to zero rather than a sheer.

- The steering system was removed and function tested on 12/07/2026 and found serviceable, with no sign of any pre-existing defect.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Section 4.2 of Waterways Aids to Navigation Code WAN-2 applied only to aids on a declared shipping channel, and the Weerona Channel was not one. Section 4.2 carries an availability target of 99.0 per cent, a duty to record availability, and a duty to notify a change of character, position or range by notice to mariners. Because it did not apply, section 4.3 governed the channel's aids and required only that the responsible authority could demonstrate an inspection interval it had set itself. (Safety issue)
2. The services agreement between Kanooka Shire Council and Kanooka Heads Port Authority set a quarterly daylight inspection and a monthly observation from the town wharf, and limited vegetation work to 50 m from the aid. It imposed no duty to notify any waterway user of a defect, a deferral or a change, and the Authority was not in breach of the agreement or of WAN-2. (Safety issue)
3. On 22/04/2026 the Operations Manager approved change record CHG-2026-014, which moved service TF18 from 16:15 to 17:35 from 01/06/2026. The change was assessed as a timetable change only, with no route risk assessment review required, although RRA-07 issue 2 restricted the channel to daylight transits and required the General Manager Operations' approval and a specific passage plan for any night transit. (Safety issue)
4. RRA-07 was not reopened and no night passage plan was written. The General Manager Operations' approval was neither sought nor given, and TF18 ran 31 times in darkness between 01/06/2026 and the occurrence.
5. The No. 4 beacon was converted to a solar lantern in 11/2025, which reduced its nominal range from 6 to 3 nautical miles. The hydrographic service was notified on 04/02/2026, but no local notice to mariners was issued and no operator using the channel was told. The operator's passage plan therefore still recorded the beacon at 6 M on 07/07/2026, and a measurement on 18/07/2026 gave a luminous range of 2.6 nm. (Safety issue)
6. Work order WO-118 was raised in 09/2023 to trim the mangrove canopy on Weerona Islet that screened the No. 4 beacon from the northern approach. It was deferred as unfunded at the 2023, 2024 and 2025 works rounds. The Authority's file note recorded that the islet, 240 m from the beacon, lay outside the 50 m scope of schedule 2.4, so the work competed for the general waterways vote. By 03/2026, WO-176 recorded the sightline as fully closed on the northern approach. (Safety issue)
7. The monthly check of the beacon was made from the town wharf, on the side of the beacon from which the screening could not be seen, and it recorded the beacon as operating every month from 01/2024.
8. Appendix A of the passage plan named the chart plotter as the primary means of position fixing. It named the visual transit of the No. 4 beacon with the town wharf leading lights as the secondary means, and it was written for the daylight transit only. It gave no guidance on losing the plotter and named no hold point or abort criterion. At the point in the transit where the plotter was lost, the beacon made up the whole of the secondary means. (Safety issue)

9. OPS-9 required the outgoing master to brief the relieving master on any defect, but required nothing of that brief to be written down and nothing of the relieving master afterwards. It required a non-statutory defect to be entered only at the end of the shift in which it was found, and the chart plotter was not a statutory item, although it was the primary means of position fixing. (Safety issue)

10. The forward saloon lifejacket locker could not be reached from 22 of the 34 forward saloon seats without passing behind a kiosk counter. The counter was fitted in a 2019 refit under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement, and the same counter is fitted to the operator's two other 24 m vessels. The locker position had no bearing on the grounding or its outcome. (Safety issue)

11. The chart plotter display froze at 18:07:06 during the channel transit and did not recover. Earlier that morning the unit had frozen at 10:42 and 11:17 with the same cause code 0x51 and had recovered by itself on both occasions. The manufacturer's firmware of 11/2024 addressing that cause code had never been applied, and the fault did not recur during a 46 hour bench test.

12. The check on 18/07/2026 found that the No. 4 beacon could not be seen from the channel between 0.84 nm and 0.34 nm north of it, because the sightline passed through mangrove canopy measured at 5.8 m against a lantern height of 4.2 m. It is very likely that the light was likewise not visible over that stretch on the night of the occurrence. The master began looking for the beacon at 0.47 nm and called it at 0.25 nm, 91 seconds later.

13. The master was on day nine of a twelve-day tour, working a split duty of 9 hours 10 minutes across a spread of 13 hours 40 minutes, inside the operator's limits and the state crewing determination. The recorded evidence of his helm work, of his response to the shallow alarm and of his radio work after the grounding gave no sign of fatigue on the night.

14. The tide gauge at Kanooka Heads read 0.91 m at 18:12 against a predicted 0.94 m, which left 1.41 m of water over the bed at the contact point. The vessel touched because it was 54 m outside the dredged channel, not because the water was lower than predicted.

15. At the handover at about 13:05 the outgoing master told the relieving master about the two morning freezes. Neither entered the fault in the defect book or reported it ashore, in keeping with a practice both masters described independently and volunteered, and the maintenance system held no plotter entry for 2026.

16. After the plotter froze, the transit continued at 8.4 kt with no reduction of speed, and stopping was not considered. The water was flat, there was no other traffic, and the vessel could have been stopped in 92 m at any point in the 114 seconds before it crossed the channel edge.

17. No lookout was on the bridge during the transit. The deckhand had been sent to collect fares at 18:04:14 in accordance with normal practice on the route, a matter OPS-3 did not address.

18. The master stopped both engines 27 seconds after contact and decided, within 60 seconds of the vessel coming to rest, not to attempt to refloat on the falling tide. He had the No. 2 starboard void sounded, which kept a holed vessel on the bank rather than putting a flooding compartment into deep water with 38 passengers aboard.

19. From 18:07:06 the heading increased from 195 to 200 degrees over about a minute and was then held. The vessel set west at an average of 12 m per minute and crossed the western edge of the 40 m wide dredged channel at about 18:09:00. It came to rest 74 m west of the charted centreline and 54 m outside the channel.

20. The echo sounder's shallow alarm had been set at 0.9 m under the keel since commissioning in 2019. It sounded at 18:11:48 and gave 16 seconds' warning, in which the vessel covered 58 m against a crash stop distance from 8 kt of 92 m.

21. The engine, steering pressure and rudder angle traces and the post-occurrence examination of the steering system showed no steering or propulsion defect, and the heading change was one that had been applied and held.

Causal factors

- The investigation found it virtually certain that the winter timetable moved the last service into darkness without reopening the route risk assessment. It is likely that this contributed, because a review of RRA-07 against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons and would have required a night passage plan, although such a review might have produced a plan that said no more than the daylight one. (Safety issue)
- It is virtually certain that there was no night passage plan for the channel and no hold point if the primary means of fixing was lost. It is very likely that this absence contributed, since a named hold point would have been available throughout the 114 seconds between the plotter freeze and the channel edge, although a master who did not think of stopping unprompted might not have applied a written rule under the same pressure. (Safety issue)
- It is virtually certain that the change-of-master defect brief was verbal, with nothing checking that what was said got written down. It is likely that this contributed, in that three or four freezes over a year never reached anyone who could act and the firmware fix of 11/2024 was never applied. The contrary reading was weighed and not preferred: an entry made at 13:05 would have been reviewed only at the end of that week, the vessel would have sailed TF18 either way, and on that view the handover rule is a finding about the fleet rather than a cause of this grounding. (Safety issue)
- It is virtually certain that the plotter fault was passed on verbally at the change of master and not entered in the defect book, an omission that OPS-9 permitted. It is likely that this contributed by leaving the fault invisible to anyone ashore, although a single entry made at 13:05 on a weekly review cycle would not by itself have stopped TF18 that evening.
- The chart plotter display froze during the channel transit and did not recover, which is virtually certain. This very likely contributed, since the vessel held the centreline while the plotter worked and left it within two minutes of the freeze.
- It is virtually certain that the transit continued at eight knots after the plotter froze, rather than stopping in the channel. It is very likely that this contributed, because a stop at any point in the 114 seconds before the vessel crossed the channel's western edge would have left it in the channel rather than on the bank, and nothing in the passage plan prompted that decision.
- It is virtually certain that aids to navigation on this waterway carried no availability standard and no duty to tell users anything. It is likely that this contributed, because the notification duty in section 4.2 was the one mechanism that would have put the beacon's range change in front of the operator. This conclusion is less secure than the other conclusions on contribution: nothing in the code prevented the Port Authority from notifying operators, trimming the canopy or measuring the light, and an availability target would not have detected a light that was lit but screened. (Safety issue)
- The conversion of the beacon to solar, which halved its nominal range, without any operator being told, is virtually certain. It likely contributed, because a weaker light against a brighter background is harder to acquire, but this was discounted from certainty because the light was screened entirely over the stretch where the master was looking, and a screened light is not made visible by being brighter. (Safety issue)

- Deferral of the clearing of the beacon's sightline, unfunded, for three works rounds is virtually certain from the Port Authority's own register. It very likely contributed, since the vessel left the channel inside the screened arc, although the master might not have picked the light up in that minute and a half even had it been there to see. (Safety issue)
- It is very likely that the No. 4 beacon was not visible from the channel over the half mile before the vessel left it, on a measurement made eleven days later from the inbound track at the same time of night. It is very likely that this contributed, because with the plotter frozen the beacon was the only means of fixing available, although the vessel had already begun to set west before the master started looking.
- It is virtually certain that the vessel left the dredged channel and tracked onto Boolarong Bank, and virtually certain that this contributed, the grounding following directly and only from the vessel being outside the channel.
- It is virtually certain that shoaling was not detected until sixteen seconds before contact. It is very likely that this contributed, because at the 0.9 m alarm setting and the speed the vessel was making, the warning gave less than the 92 m the vessel needed to stop, although the equipment worked exactly as it was set.

Contributory factors

- It is virtually certain that the forward saloon lifejacket locker was not reachable from most of the saloon it serves, because 22 of its 34 seats had no route to it except behind a counter fitted in a refit that triggered no re-survey. It is very likely that this increased risk, although it had no bearing on the grounding or its outcome. (Safety issue)
- The master's duty pattern, day nine of twelve on a split duty across a thirteen hour forty spread, is virtually certain as rostered. It is likely that this pattern increased risk, because it recurs on every darkness transit of the winter timetable, although the recorded evidence gave no sign of fatigue on the night and the pattern was not found to have influenced this occurrence.

Other findings

- The hypothesis that a water level below prediction reduced the depth over Boolarong Bank was examined and ruled out. The gauge read 0.91 m at 18:12 against a predicted 0.94 m.
- The hypothesis that a steering or propulsion failure caused the vessel to sheer off track was examined and ruled out. The recorded engine, steering and rudder traces showed no fault, and every component of the steering system was found serviceable on examination.
- The master's decision to stop the engines and make no attempt to refloat on a falling tide was examined and found to have reduced risk, because the breach was in the compartment on which the vessel was resting.

The investigation also examined whether new foreshore floodlighting behind the beacon reduced the master's ability to pick it up, and could not settle the matter. Fourteen unshielded LED floodlights, commissioned on 22/05/2026, lie behind the beacon within 4 degrees of the same bearing as seen from 0.5 nm north of it. Both masters independently described the beacon as lost against the town and foreshore lighting, and no assessment of the development against the channel's aids was required or made. Against this, the one position at which the alignment was measured lies inside the screened arc, where the beacon would not have been visible against a dark background either. The master also acquired the light once clear of the screening, with the floodlights unchanged. No measurement exists of what could be seen on the night, and nothing establishes a detection threshold for a 0.4 second green flash against that background, which may

not be determinable while the canopy screens the light. The evidence is evenly balanced and no finding is made on it.

Safety action and recommendations

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

WAN-2 applies three duties only to aids on a declared shipping channel. These are an availability target of 99.0 per cent, a duty to record availability, and a duty to notify any change of character, position or range by notice to mariners. The Weerona Channel is not a declared shipping channel. Its beacons therefore fall under section 4.3, which requires only that the responsible authority can demonstrate an inspection interval it has set for itself. The services agreement that sets that interval imposes no duty to tell any waterway user of a defect, a deferral or a change. It is virtually certain that this gap existed. The Port Authority was not in breach of either instrument.

The safety significance of the issue lies in what it left unreported on a channel 40 m wide used by a 96-passenger ferry. It is likely that the absence of a notification duty contributed to the grounding. A notice to mariners was the one mechanism identified that would have put the halving of the No. 4 beacon's nominal range in front of the operator. The operator's passage plan still carried the old figure on the night. This finding is held with less confidence than the other causal links in this report. Nothing in the code prevented the Authority from notifying operators, measuring the light or clearing the mangroves. An availability target would also not have detected a lantern that was lit but could not be seen.

No safety action has been recorded in response to this issue, and this report makes no recommendation on it.

Beacon converted to solar, nominal range halved, and no operator was told

In 11/2025 the No. 4 beacon was converted to a solar lantern with half the nominal range of the lantern it replaced, and no operator using the channel was told. On the night of the grounding, the operator's passage plan appendix still recorded the beacon at 6 M. By then the lantern had been a 3 M unit for eight months, and its luminous range was later measured at 2.6 nm. The master was therefore navigating to a written plan that described a light more than twice as strong as the one in front of him. That light also stood against a background of town and foreshore lighting added in the same period.

It is virtually certain that the condition existed. It is likely that it contributed to the occurrence, because a weaker light against a brighter background is harder to acquire. The assessment goes no further than likely because, over the stretch where the master was looking, the light was screened entirely, and a brighter light would not have been seen there either. This is a causal factor.

Safety action taken

On 22/07/2026 Kanooka Shire Council issued local notice to waterway users LNM-2026-14. The notice gave the beacon's solar conversion and its 3 M nominal range. The Port Authority's aids to navigation register was also made available to every commercial operator using the channel. Tarrakin Ferries has reissued the Weerona Channel passage plan appendix as issue 3, with the corrected range. This action is recorded as in progress.

The action corrects the range information held by operators. It answers this issue in part, because it does not of itself create a continuing duty to tell users of future changes. That gap is the subject of the wider safety issue concerning the aids to navigation regime. [NOT ESTABLISHED: whether the chart correction pending with the hydrographic service has since been published.]

Safety recommendations

No further safety action has been recorded for this issue, and no recommendation is made.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Mangrove regrowth on Weerona Islet had screened the No. 4 beacon from the northern approach since at least 09/2023. In that month, work order WO-118 recommended trimming the canopy to 2 m on a three-yearly cycle, at an estimated cost of \$18,500. The work was deferred as unfunded at the 2023, 2024 and 2025 works rounds. The islet lies 240 m from the beacon, outside the 50 m scope of schedule 2.4, so the work competed for the general waterways vote rather than the aids to navigation vote. By 03/2026 the Authority's register recorded the sightline as "now fully closed on the northern approach". This deferral was a causal factor. It left the beacon absent from view over the stretch of channel in which the vessel left it.

Safety action taken

The Harbour Master, Kanooka Heads, funded WO-176 from the 2026 waterways contingency. The canopy was trimmed to 2 m on 24 July 2026, restoring the beacon's sightline over the northern approach. A three-yearly trimming cycle has been programmed, and the sightline was re-measured from the inbound track on completion. This answers the screening at the No. 4 beacon in full. It does not change the scope rule that allowed the work to be deferred.

Safety recommendation

The 50 m vegetation scope places work needed to keep an aid visible outside the aids to navigation vote whenever the obstruction lies further from the structure. It is recommended that [NOT ESTABLISHED: organisation to which the recommendation is addressed] ensure that the aids to navigation services agreement replaces that scope with a sightline test. Work needed to keep an aid visible from the water it serves would then be funded as an aid to navigation, whatever its distance from the structure.

Winter timetable moved the last service into darkness without reopening the route risk assessment

The route risk assessment for the Weerona Channel, RRA-07 issue 2, restricted the channel to daylight transits. It required the General Manager Operations' approval and a specific passage plan for any night transit. Change record CHG-2026-014 was approved by the Operations Manager on 22/04/2026. It moved service TF18 from 16:15 to 17:35 from 01/06/2026, assessed the change as a timetable change only, and recorded that no review of the route risk assessment was required.

It is virtually certain that the change was made in this way. RRA-07 was not reopened and no night passage plan was written. The approval the restriction called for was neither sought nor given. It is likely that this contributed to the absence of a night passage plan for the channel, and of any hold point should the primary means of fixing be lost. It was a causal factor in the occurrence.

The issue matters beyond this transit. The change sent a 96-passenger vessel through a 40 m channel in darkness 31 times before the occurrence, contrary to a restriction that was still in force. The process asked whether the change introduced anything new rather than whether the existing assessment still held. It also allowed the change to be approved below the level the restriction named.

No safety action has been recorded in response to this issue, and no safety recommendation is made.

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The operator's passage plan for the Weerona Channel was written for the daylight transit and stated that night transits were not scheduled. It named the chart plotter as the primary means of position fixing. The secondary means was the visual transit of the No. 4 beacon with the town wharf leading lights. The plan gave no guidance on losing the plotter during the transit and named no abort or hold point. It is virtually certain that no night passage plan existed when the winter timetable moved the last service into darkness.

The absence of a hold point very likely contributed to the transit continuing at 8 kt after the plotter froze, and it was a causal factor in this occurrence. The master had 114 seconds between the freeze and crossing the channel edge, on flat water with no other traffic. A named hold point would have been available throughout that time. Both masters interviewed said independently that stopping was the right response and that nothing told them to stop.

After the occurrence, the General Manager Operations suspended evening Muntari services. [NOT ESTABLISHED: whether this suspension remains in force, and whether it was taken in response to this safety issue.] The suspension does not supply the missing plan, so at most it answers the issue in part.

It is recommended that the operator write a night passage plan for the Weerona Channel before any further transit in darkness. The plan should name a hold point under which the loss of the primary means of position fixing inside the channel requires the way to be taken off the vessel until the position is re-established by another means.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

The operator's Operations Manual OPS-9, revision 4, required immediate reporting only of a defect that rendered the vessel unseaworthy or affected a statutory item. The chart plotter was not a statutory item on a class 1D vessel on this route, although the passage plan named it as the primary means of position fixing. All other defects were to be entered in the vessel defect book at the end of the shift in which they were found, which gave no effective trigger for a fault that came and went. Under OPS-9 9.3 the outgoing master briefed the relieving master on the state of the vessel, but nothing about that brief had to be written down and nothing was required of the relieving master afterwards. Both masters independently described the same understanding: the book was for what was broken at handover, and what had come good was spoken about. This explains why the plotter fault was passed on verbally rather than booked. It does not excuse the omission.

It is likely that this issue contributed to the grounding, and it is a causal factor. Three or four earlier freezes over the preceding year never reached anyone ashore who could act. As a result, a firmware fix issued in 11/2024 was never applied, and the master met the fault with no reason to treat it as more than a nuisance.

A contrary reading was also considered. An entry made at 13:05 on the day would have been reviewed by the Operations Manager at the end of that week, and the vessel would have sailed TF18 either way. On that reading the rule is a real finding about the fleet but not a cause of this grounding.

No safety action has been recorded in response to this issue, and no recommendation is made.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

The lifejacket locker in the forward saloon could not be reached from most of the saloon it served. A survey of the vessel on 16 July 2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the kiosk counter. With people standing at the counter, the 0.6 m inboard gap was the only route. The muster placard directed passengers to a locker they could not see from their seats. The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted in a 2019 interior refit under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

It is unlikely that this arrangement influenced the occurrence or its outcome. All 38 passengers were in lifejackets within about eight minutes and ashore within 74 minutes. It is nevertheless very likely that it increased risk. On this occasion the vessel was aground, upright and stable, and the muster was unhurried. In a fire, a night evacuation into the water, or with the deckhand injured in the impact, a locker that passengers cannot reach without crew help would leave the saloon waiting to be equipped rather than equipping itself. The underlying weakness lay in the modification route. It allowed an interior refit to change a safety equipment arrangement without any check of whether it had done so.

No safety action has been recorded in response to this safety issue, and this report makes no safety recommendation about it.

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